

→ Linblad Master Equation

$$|\psi(t+dt)\rangle = \left(\mathbb{I} - \frac{i}{\hbar} \hat{H} \cdot dt \right) |\psi(t)\rangle$$

$$\rho(t+dt) = \rho(t) - \frac{i}{\hbar} \cdot dt [\hat{H}, \rho(t)]$$

$$\rho(t+dt) = \mathcal{E}_{dt}(\rho(t))$$

$$\mathcal{E}_{dt} = \text{Id} + dt \cdot \underbrace{\mathcal{L}}_{\text{Linbladian / Liouvillian}}$$

$$\frac{\rho(t+dt) - \rho(t)}{dt} = \frac{\partial \rho}{\partial t} = \dot{\rho} = \mathcal{L}(\rho) = -\frac{i}{\hbar} [\hat{H}, \rho(t)]$$

Non-Markovian Evolution

Notes on
GitHub
utsavkaria-tech